

InternNova – Week 1 Assignment

Python Fundamentals

Task 1: Python Basics

Marks: 10

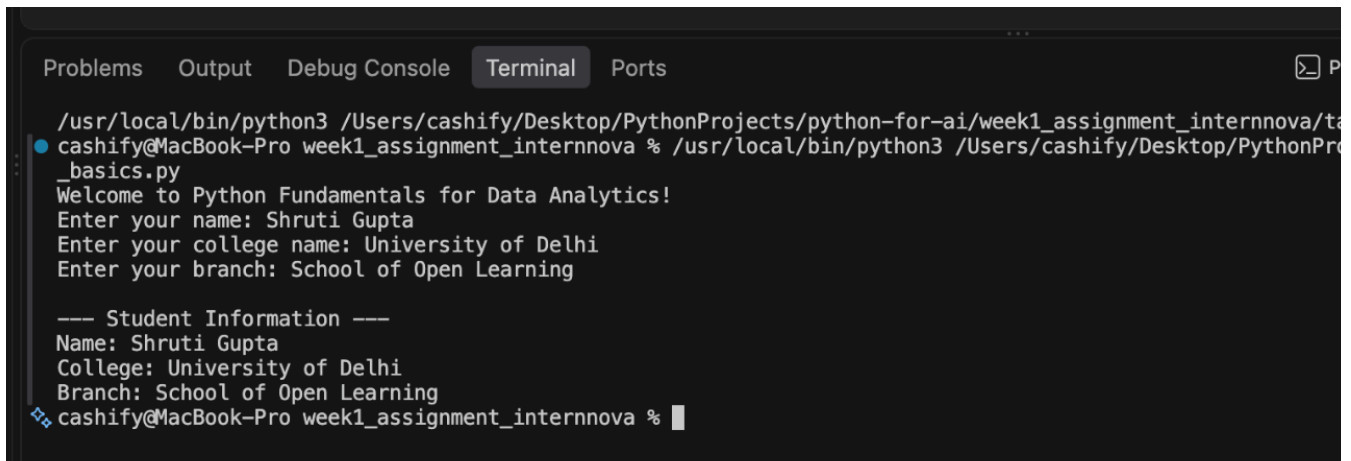
1. Brief Explanation

This program prints a welcome message and takes the student's name, college name, and branch as input. It then displays the entered information in a formatted output.

2. Python Source Code

```
print("Welcome to Python Fundamentals for Data  
Analytics!")  
  
name = input("Enter your name: ")  
college = input("Enter your college name: ")  
branch = input("Enter your branch: ")  
  
print("\n--- Student Information ---")  
print(f"Name: {name}")  
print(f"College: {college}")  
print(f"Branch: {branch}")
```

3. Program Output



```
Problems Output Debug Console Terminal Ports
/usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/ta
cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonPro
_basics.py
Welcome to Python Fundamentals for Data Analytics!
Enter your name: Shruti Gupta
Enter your college name: University of Delhi
Enter your branch: School of Open Learning

--- Student Information ---
Name: Shruti Gupta
College: University of Delhi
Branch: School of Open Learning
cashify@MacBook-Pro week1_assignment_internnova %
```

4. Conclusion

This task helped me understand the basic use of Python input and output statements. I learned how to take information from the user and display it in a clear and formatted manner. It also provided a basic understanding of Python programming and user interaction.

Task 2: Variables & Data Types

Marks: 10

1. Brief Explanation

This program demonstrates four basic Python data types: integer, float, string, and boolean. It stores values of each type in variables and uses the `type()` function to identify and display the data type of each value. A loop is used to print the value along with its corresponding data type in a clear format.

2. Python Source Code

```
integer_value = 25
float_value = 3.14
string_value = "Python"
boolean_value = True

values = [
    ("Integer", integer_value),
    ("Float", float_value),
```

```

    ("String", string_value),
    ("Boolean", boolean_value),
]

for label, value in values:
    print(f"{label}: {value} | Type: {type(value).__name__}")

```

3. Program Output

```

/usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task2_data_types.py
cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task2_data_types.py
Integer: 25 | Type: int
Float: 3.14 | Type: float
String: Python | Type: str
Boolean: True | Type: bool
cashify@MacBook-Pro week1_assignment_internnova %

```

4. Conclusion

This task helped me understand the different basic data types in Python, including integers, floats, strings, and booleans. I also learned how to use the `type()` function to identify the data type of a value and how to use a loop to display the information in a clear and organized format.

Task 3: Operators

Marks: 10

1. Brief Explanation

This program works as a simple calculator that takes two numbers as input from the user. It performs five basic arithmetic operations: addition, subtraction, multiplication, division, and modulus. An `if-else` condition is used to handle division by zero and display an appropriate message instead of causing an error.

2. Python Source Code

```

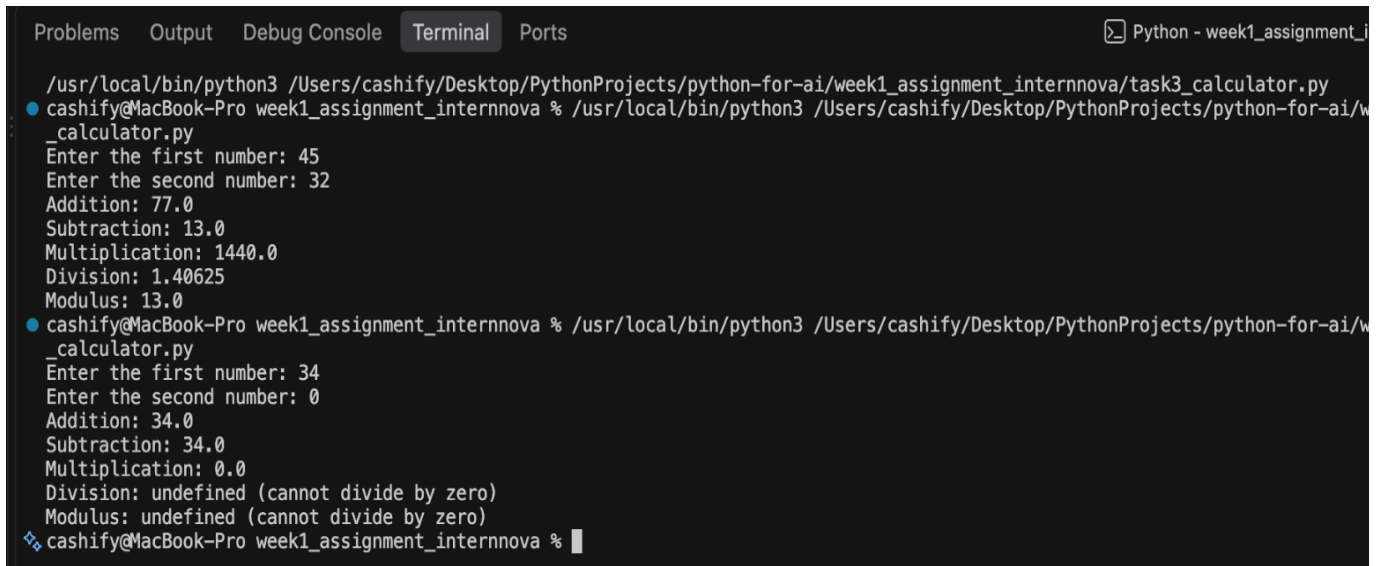
first_number = float(input("Enter the first number: "))
second_number = float(input("Enter the second number: "))

print(f"Addition: {first_number + second_number}")
print(f"Subtraction: {first_number - second_number}")
print(f"Multiplication: {first_number * second_number}")

if second_number == 0:
    print("Division: undefined (cannot divide by zero)")
    print("Modulus: undefined (cannot divide by zero)")
else:
    print(f"Division: {first_number / second_number}")
    print(f"Modulus: {first_number % second_number}")

```

3. Program Output



```

Problems  Output  Debug Console  Terminal  Ports
Python - week1_assignment_i

/usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task3_calculator.py
● cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/w
_calculator.py
Enter the first number: 45
Enter the second number: 32
Addition: 77.0
Subtraction: 13.0
Multiplication: 1440.0
Division: 1.40625
Modulus: 13.0
● cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/w
_calculator.py
Enter the first number: 34
Enter the second number: 0
Addition: 34.0
Subtraction: 34.0
Multiplication: 0.0
Division: undefined (cannot divide by zero)
Modulus: undefined (cannot divide by zero)
❖ cashify@MacBook-Pro week1_assignment_internnova %

```

4. Conclusion

This task helped me understand how arithmetic operators are used in Python to perform calculations. I also learned how to take numerical input from the user and use conditional statements to handle special cases such as division by zero.

Task 4: Conditional Statements

Marks: 10

1. Brief Explanation

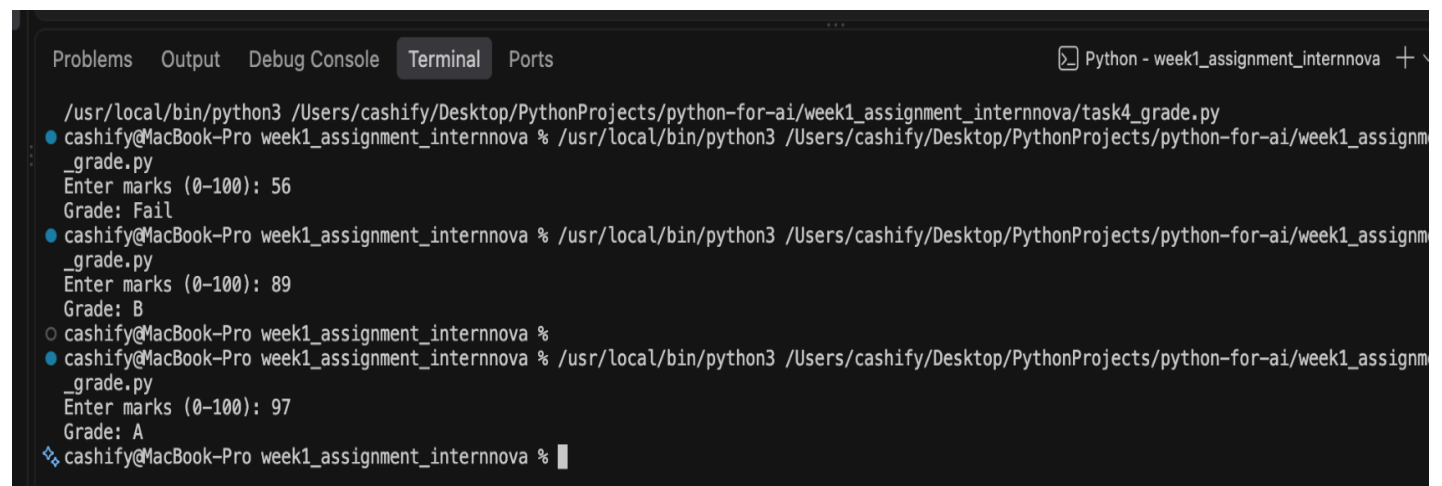
This program takes marks as input from the user and assigns a grade based on the given marks range. It uses **if**, **elif**, and **else** conditional statements to check different conditions. The program also validates the marks to ensure that the entered value is between 0 and 100.

2. Python Source Code

```
marks = float(input("Enter marks (0-100): "))

if marks < 0 or marks > 100:
    print("Invalid marks. Enter a value from 0 to 100.")
elif marks >= 90:
    print("Grade: A")
elif marks >= 75:
    print("Grade: B")
elif marks >= 60:
    print("Grade: C")
else:
    print("Grade: Fail")
```

3. Program Output



```
Problems Output Debug Console Terminal Ports Python - week1_assignment_internnova +
/usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task4_grade.py
cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task4_grade.py
Enter marks (0-100): 56
Grade: Fail
cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task4_grade.py
Enter marks (0-100): 89
Grade: B
cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task4_grade.py
Enter marks (0-100): 97
Grade: A
cashify@MacBook-Pro week1_assignment_internnova %
```

4. Conclusion

This task helped me understand how conditional statements can be used in Python to make decisions based on user input. I learned how to apply **if**, **elif**, and **else** to assign grades and how to handle invalid input using conditions.

Task 5: Loops

Marks: 10

1. Brief Explanation

This program demonstrates the use of **for** and **while** loops in Python. It uses a **for** loop to print numbers from 1 to 20 and to generate the multiplication table of a user-entered number. A **while** loop is used to print even numbers from 1 to 50. The program shows how loops can be used to repeat instructions efficiently.

2. Python Source Code

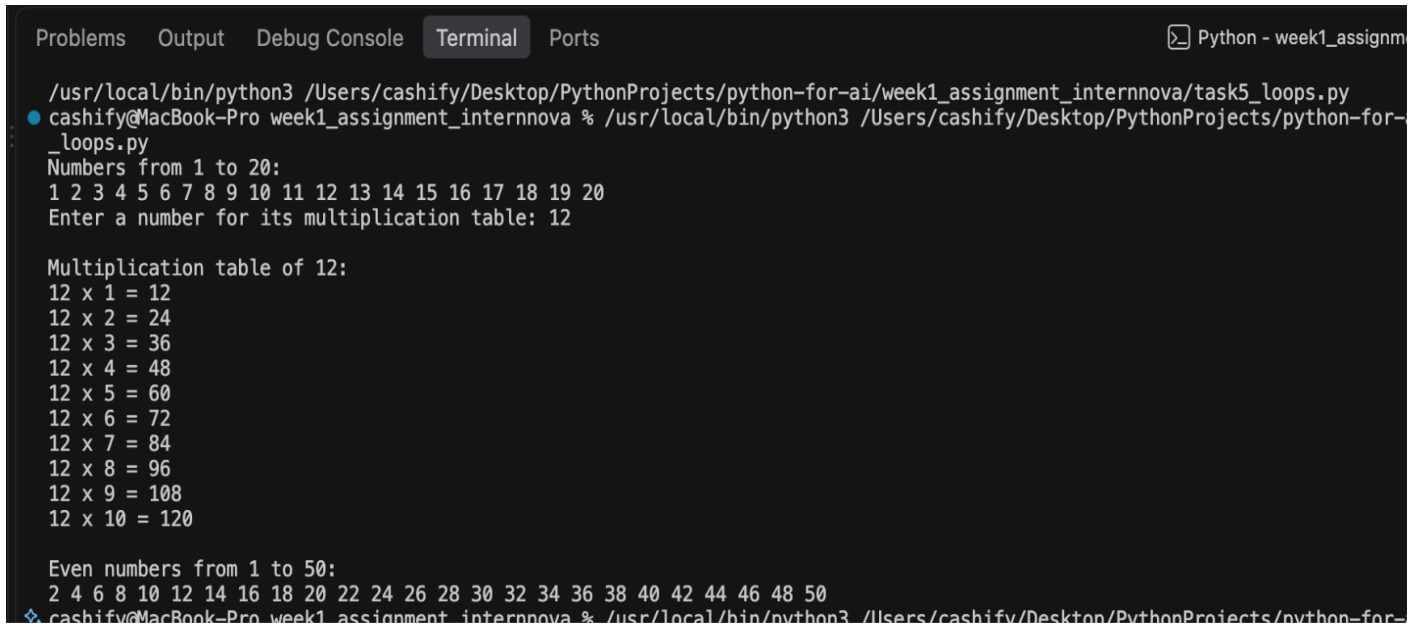
```
print("Numbers from 1 to 20:")
for number in range(1, 21):
    print(number, end=" ")
print()

number = int(input("Enter a number for its multiplication
table: "))
print(f"\nMultiplication table of {number}:")
for multiplier in range(1, 11):
    print(f"{number} x {multiplier} = {number *
multiplier}")

print("\nEven numbers from 1 to 50:")
even_number = 2
while even_number <= 50:
    print(even_number, end=" ")
    even_number += 2
```

```
print()
```

3. Program Output



```
Problems Output Debug Console Terminal Ports Python - week1_assignm
/usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task5_loops.py
cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-
_loops.py
Numbers from 1 to 20:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Enter a number for its multiplication table: 12

Multiplication table of 12:
12 x 1 = 12
12 x 2 = 24
12 x 3 = 36
12 x 4 = 48
12 x 5 = 60
12 x 6 = 72
12 x 7 = 84
12 x 8 = 96
12 x 9 = 108
12 x 10 = 120

Even numbers from 1 to 50:
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50
cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-
```

4. Conclusion

This task helped me understand how **for** and **while** loops are used in Python for repetitive tasks. I learned how to generate sequences of numbers, create a multiplication table, and print even numbers using loops.

Task 6: Functions

Marks: 10

1. Brief Explanation

This program demonstrates the use of user-defined functions in Python. It defines one function to calculate the square of a number and another function to calculate the average of three numbers. The program takes the required numbers as input from the user, calls the functions, and displays the calculated results.

2. Python Source Code

```
def square(number):
    return number * number

def average(first_number, second_number, third_number):
```

```

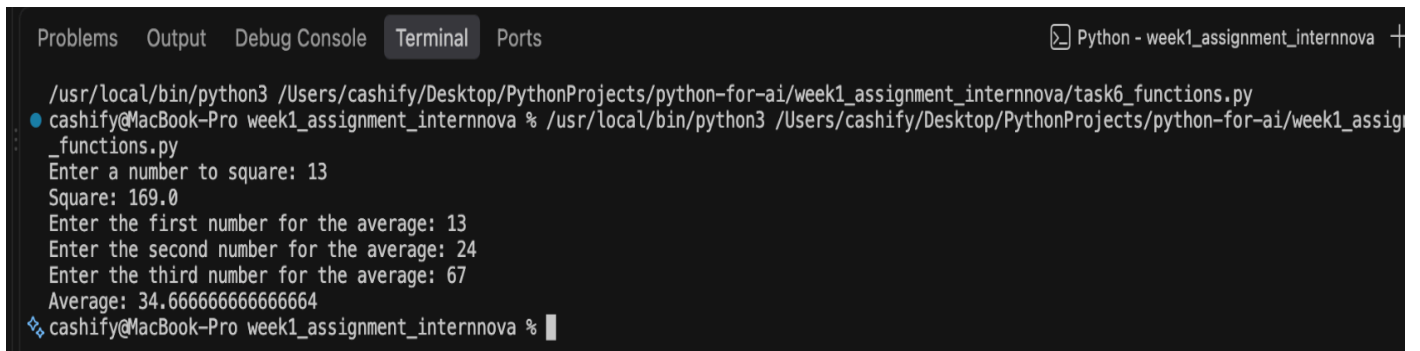
    return (first_number + second_number + third_number) /
3

number = float(input("Enter a number to square: "))
print(f"Square: {square(number)}")

numbers = [
    float(input("Enter the first number for the average:
")),
    float(input("Enter the second number for the average:
")),
    float(input("Enter the third number for the average:
")),
]
print(f"Average: {average(*numbers)}")

```

3. Program Output



```

Problems  Output  Debug Console  Terminal  Ports
Python - week1_assignment_internnova +

/usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task6_functions.py
cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task6_functions.py
Enter a number to square: 13
Square: 169.0
Enter the first number for the average: 13
Enter the second number for the average: 24
Enter the third number for the average: 67
Average: 34.666666666666664
cashify@MacBook-Pro week1_assignment_internnova %

```

4. Conclusion

This task helped me understand how user-defined functions are created and used in Python. I learned how to pass values to functions, perform calculations, and return results. It also showed how functions can make a program more organized and reusable.

Task 7: Strings & Collections

Marks: 15

1. Brief Explanation

This program demonstrates different string and collection operations in Python. It uses `upper()`, `lower()`, `replace()`, and `find()` for string manipulation. It also demonstrates `append()`, `remove()`, and `sort()` on a list. A tuple is created and accessed using indexing, a dictionary is used to store student information, and a set is modified using `add()` and `remove()`.

2. Python Source Code

```
text = "Python makes data analysis powerful"
print(f"Original string: {text}")
print(f"Uppercase: {text.upper()}")
print(f"Lowercase: {text.lower()}")
print(f"Replace: {text.replace('powerful',
'accessible')}")
print(f"Position of 'data': {text.find('data')}")

numbers = [30, 10, 20]
numbers.append(40)
numbers.remove(10)
numbers.sort()
print(f"\nList after append, remove, and sort: {numbers}")

student_tuple = ("Asha", "Computer Science", 21)
print(f"Tuple: {student_tuple}")
print(f"Tuple item at index 0: {student_tuple[0]}")

student = {"name": "Asha", "age": 21, "branch": "Computer
Science"}
print(f"Student dictionary: {student}")
```

```
skills = {"Python", "SQL"}
skills.add("Excel")
skills.remove("SQL")
print(f"Set after add and remove: {skills}")
```

3. Program Output

```
Python - week1_assignment_internnova

/usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task7_strings_collections.py
● cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_strings_collections.py
Original string: Python makes data analysis powerful
Uppercase: PYTHON MAKES DATA ANALYSIS POWERFUL
Lowercase: python makes data analysis powerful
Replace: Python makes data analysis accessible
Position of 'data': 13

List after append, remove, and sort: [20, 30, 40]
Tuple: ('Asha', 'Computer Science', 21)
Tuple item at index 0: Asha
Student dictionary: {'name': 'Asha', 'age': 21, 'branch': 'Computer Science'}
Set after add and remove: {'Python', 'Excel'}
❖ cashify@MacBook-Pro week1_assignment_internnova %
```

4. Conclusion

This task helped me understand how different collection types and string methods are used in Python. I learned how to manipulate strings and perform basic operations on lists, tuples, dictionaries, and sets. These concepts are useful for storing, organizing, and working with data efficiently.

Task 8: Basic File Handling

Marks: 10

1. Brief Explanation

This program demonstrates basic file handling in Python. It creates a text file named `introduction.txt` and writes a short introduction into the file. The program then opens the file in read mode, reads its contents, and displays the information on the screen.

2. Python Source Code

```
file_name = "introduction.txt"
introduction = (
```

```
"Hello, my name is Shruti. I am learning Python for
Data Analytics.\n"

"Python helps me work with data and solve problems."
)

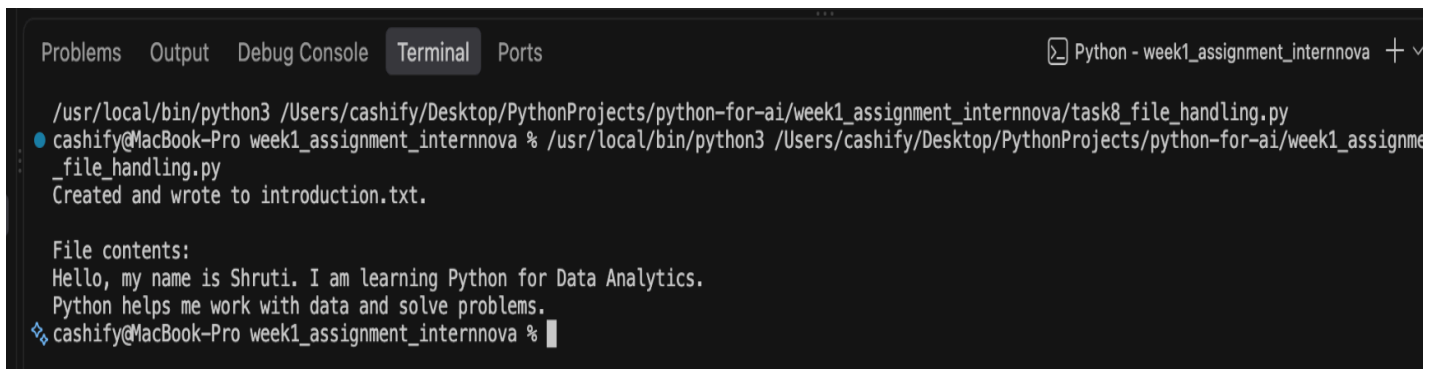
with open(file_name, "w", encoding="utf-8") as file:
    file.write(introduction)

print(f"Created and wrote to {file_name}.")

with open(file_name, "r", encoding="utf-8") as file:
    contents = file.read()

print("\nFile contents:")
print(contents)
```

3. Program Output



```
Problems Output Debug Console Terminal Ports Python - week1_assignment_internnova + v
/usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task8_file_handling.py
cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/task8_file_handling.py
Created and wrote to introduction.txt.

File contents:
Hello, my name is Shruti. I am learning Python for Data Analytics.
Python helps me work with data and solve problems.
cashify@MacBook-Pro week1_assignment_internnova %
```

4. Conclusion

This task helped me understand the basic process of creating, writing, reading, and displaying the contents of a text file in Python. I also learned how to use the **with open()** statement for handling files efficiently.

Task 9: Mini Python Project

Marks: 15

1. Brief Explanation

This project is a menu-driven Student Record Management System developed using Python. It stores student details in a list of dictionaries and provides options to add, display, search, and delete student records. Separate functions are used for each operation, making the program organized and easy to manage.

2. Python Source Code

```
students = []

def add_student():
    name = input("Enter student name: ").strip()
    age = input("Enter student age: ").strip()
    branch = input("Enter student branch: ").strip()
    students.append({"name": name, "age": age, "branch":
branch})
    print("Student added successfully.")

def display_students():
    if not students:
        print("No student records found.")
        return

    print("\n--- Student Records ---")
    for index, student in enumerate(students, start=1):
        print(
            f"{index}. Name: {student['name']}, "
```

```

        f"Age: {student['age']}, Branch:
{student['branch']}"
    )

def search_student():
    search_name = input("Enter the name to search:
").strip().lower()
    matches = [
        student for student in students
        if search_name in student["name"].lower()
    ]

    if not matches:
        print("Student not found.")
        return

    for student in matches:
        print(
            f"Name: {student['name']}, "
            f"Age: {student['age']}, Branch:
{student['branch']}"
        )

def delete_student():
    delete_name = input("Enter the name to delete:
").strip().lower()
    for index, student in enumerate(students):
        if student["name"].lower() == delete_name:

```

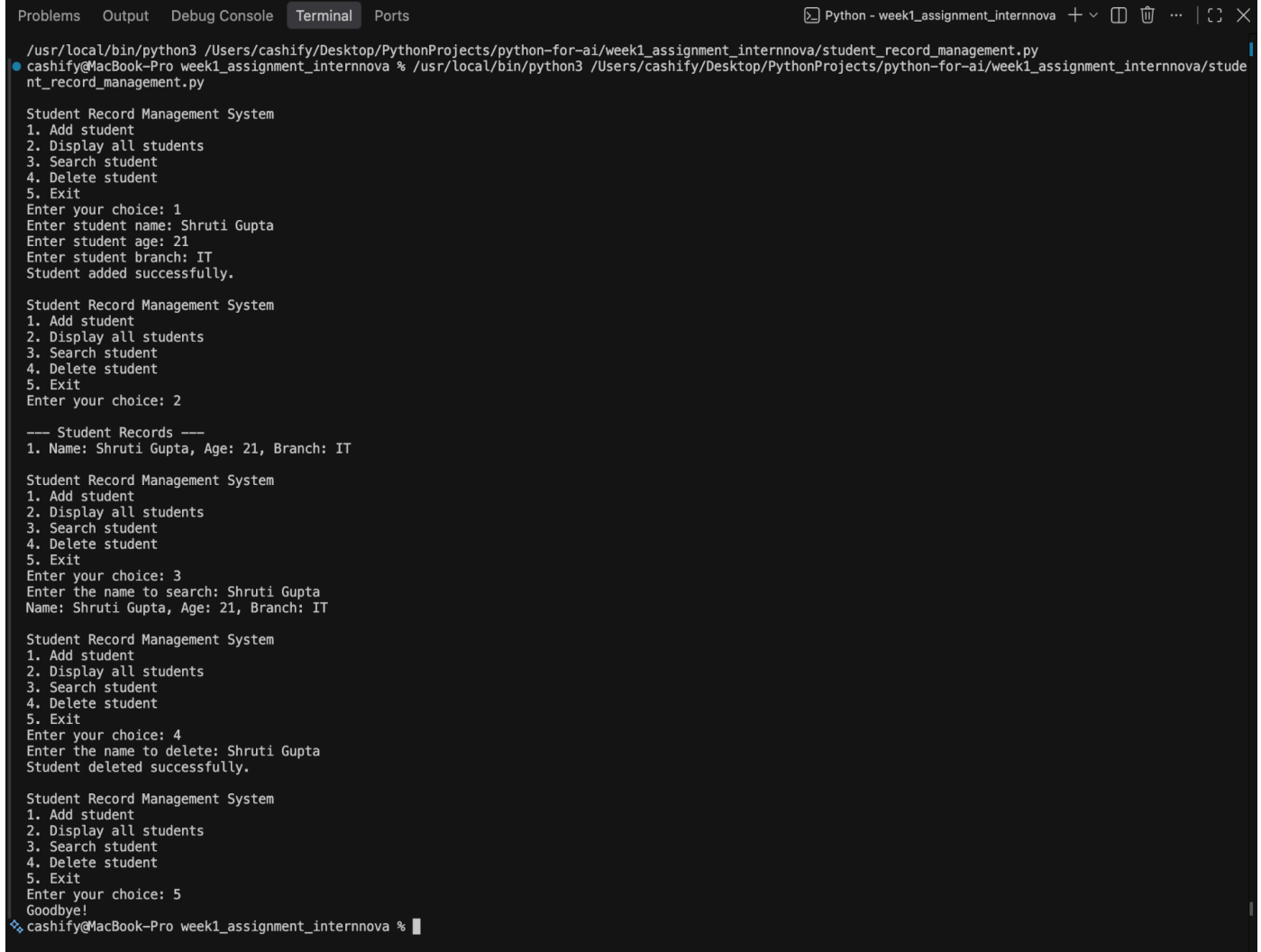
```
        students.pop(index)
        print("Student deleted successfully.")
        return
    print("Student not found.")

def main():
    while True:
        print("\nStudent Record Management System")
        print("1. Add student")
        print("2. Display all students")
        print("3. Search student")
        print("4. Delete student")
        print("5. Exit")
        choice = input("Enter your choice: ").strip()

        if choice == "1":
            add_student()
        elif choice == "2":
            display_students()
        elif choice == "3":
            search_student()
        elif choice == "4":
            delete_student()
        elif choice == "5":
            print("Goodbye!")
            break
        else:
            print("Invalid choice. Please select 1 to 5.")
```

```
if __name__ == "__main__":  
    main()
```

3. Program Output



```
Python - week1_assignment_internnova + - [ ] [ ] ... | [ ] [ ] X  
/usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/student_record_management.py  
cashify@MacBook-Pro week1_assignment_internnova % /usr/local/bin/python3 /Users/cashify/Desktop/PythonProjects/python-for-ai/week1_assignment_internnova/stude  
nt_record_management.py  
  
Student Record Management System  
1. Add student  
2. Display all students  
3. Search student  
4. Delete student  
5. Exit  
Enter your choice: 1  
Enter student name: Shruti Gupta  
Enter student age: 21  
Enter student branch: IT  
Student added successfully.  
  
Student Record Management System  
1. Add student  
2. Display all students  
3. Search student  
4. Delete student  
5. Exit  
Enter your choice: 2  
  
--- Student Records ---  
1. Name: Shruti Gupta, Age: 21, Branch: IT  
  
Student Record Management System  
1. Add student  
2. Display all students  
3. Search student  
4. Delete student  
5. Exit  
Enter your choice: 3  
Enter the name to search: Shruti Gupta  
Name: Shruti Gupta, Age: 21, Branch: IT  
  
Student Record Management System  
1. Add student  
2. Display all students  
3. Search student  
4. Delete student  
5. Exit  
Enter your choice: 4  
Enter the name to delete: Shruti Gupta  
Student deleted successfully.  
  
Student Record Management System  
1. Add student  
2. Display all students  
3. Search student  
4. Delete student  
5. Exit  
Enter your choice: 5  
Goodbye!  
cashify@MacBook-Pro week1_assignment_internnova %
```

4. Conclusion

This project helped me apply the Python concepts learned throughout the week in a practical program. I used lists, dictionaries, functions, loops, conditional statements, and user input to create a simple student record management system. It improved my understanding of how different Python concepts can be combined to solve a practical problem.